



Fiber Products of Affine Schemes

Understanding $\operatorname{Spec} \mathbb{Z}[x]$ as a family of curves

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| What are all the prime ideals of $\mathbb{Z}[x]$?

The ring $\mathbb{Z}[x]$ consists of all polynomials with integer coefficients.

Question. Describe all prime ideals $\mathfrak{p} \subset \mathbb{Z}[x]$.

Strategy. Given $\mathfrak{p} \in \operatorname{Spec} \mathbb{Z}[x]$, look at the *contraction* to the base:

$$\mathfrak{p} \cap \mathbb{Z} \in \operatorname{Spec} \mathbb{Z}.$$

Since $\mathbb{Z}$ is a PID, every prime is either (0) or (p) for a prime number p . This gives a natural *two-case split*.

| The case split

Case 1: $\mathfrak{p} \cap \mathbb{Z} = (0)$.

Invert $\mathbb{Z} \setminus \{0\}$: the prime $\mathfrak{p}$ extends to a prime of $\mathbb{Q}[x]$. Since $\mathbb{Q}[x]$ is a PID, the primes are:

- (0) — the zero ideal;
- $(q(x))$ for $q \in \mathbb{Z}[x]$ irreducible and primitive.

Case 2: $\mathfrak{p} \cap \mathbb{Z} = (p)$.

Reduce mod p : the prime $\mathfrak{p}$ corresponds to a prime of $\mathbb{F}_p[x] \cong \mathbb{Z}[x]/(p)$. Since $\mathbb{F}_p[x]$ is a PID, the primes are:

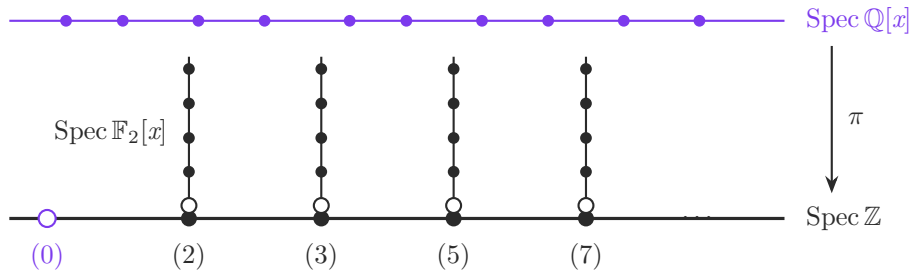
- (p) — the zero ideal of $\mathbb{F}_p[x]$, lifted back to $\mathbb{Z}[x]$;
- $(p, q(x))$ for q irreducible over $\mathbb{F}_p$.

Four types of primes in $\operatorname{Spec} \mathbb{Z}[x]$

Type	Prime	Lies over	Character
1	(0)	(0)	Generic point of $\operatorname{Spec} \mathbb{Z}[x]$
2	$(q(x))$, q irred. primitive	(0)	Closed points of the $\mathbb{Q}$ -fiber
3	(p) , p prime	(p)	Generic point of fiber over p
4	$(p, q(x))$, q irred. mod p	(p)	Closed points of fiber over p

The pattern. The primes organize into natural *fibers over* $\operatorname{Spec} \mathbb{Z}$. But what *is* a fiber, as a scheme?

| $\text{Spec } \mathbb{Z}[x]$ as a family of lines



Each vertical slice is a copy of $\text{Spec } \mathbb{F}_p[x]$; the top line is $\text{Spec } \mathbb{Q}[x]$. These fibers are visible as *sets* — but do they carry a natural *scheme structure*?

| Affine schemes — a minimal introduction

A **commutative ring** A (e.g. $\mathbb{Z}$, $k[x]$, $\mathbb{Z}[x]$) determines a geometric object:

$$\mathrm{Spec} A := \{\text{prime ideals } \mathfrak{p} \subset A\}.$$

- A prime ideal $\mathfrak{p}$ is *closed* if it is maximal; otherwise it is a *generic point*, whose closure contains all primes that contain it.
- **Example:** $\mathrm{Spec} \mathbb{Z}$ has one generic point (0) and one closed point (p) for each prime number p .
- **Example:** $\mathrm{Spec} \mathbb{F}_p[x]$ has one generic point (0) and one closed point $(q(x))$ for each monic irreducible $q \in \mathbb{F}_p[x]$ — this is the *affine line* over $\mathbb{F}_p$.

| Ring homomorphisms and morphisms of schemes

A ring homomorphism $\varphi : B \rightarrow A$ induces a map of schemes going *the other way*:

$$f = \operatorname{Spec} \varphi : \operatorname{Spec} A \longrightarrow \operatorname{Spec} B, \quad \mathfrak{q} \longmapsto \varphi^{-1}(\mathfrak{q}).$$

One checks that if $\mathfrak{q}$ is prime in A then $\varphi^{-1}(\mathfrak{q})$ is prime in B .

Our running example. The inclusion $\iota : \mathbb{Z} \hookrightarrow \mathbb{Z}[x]$ gives the projection

$$\pi = \operatorname{Spec} \iota : \operatorname{Spec} \mathbb{Z}[x] \longrightarrow \operatorname{Spec} \mathbb{Z}, \quad \mathfrak{p} \longmapsto \mathfrak{p} \cap \mathbb{Z}.$$

This is precisely the contraction $\mathfrak{p} \cap \mathbb{Z}$ from our case split — now seen as a morphism of schemes.

Localization

Let A be a ring and $S \subseteq A$ a *multiplicative set* ($1 \in S$, closed under multiplication). The **localization** $S^{-1}A$ is the ring of formal fractions

$$\frac{a}{s}, \quad a \in A, \quad s \in S,$$

where $\frac{a}{s} = \frac{b}{t}$ iff $u(at - bs) = 0$ for some $u \in S$.

Key examples.

- $A = \mathbb{Z}$, $S = \mathbb{Z} \setminus \{0\}$: $S^{-1}\mathbb{Z} = \mathbb{Q}$.
- $A = \mathbb{Z}$, $S = \mathbb{Z} \setminus (p)$: $S^{-1}\mathbb{Z} = \mathbb{Z}_{(p)}$ (integers with denominators coprime to p).
- A an *integral domain*, $S = A \setminus \{0\}$: $S^{-1}A = \text{Frac}(A)$, the fraction field.

The last example is what we need next: the *residue field* at a prime will be the fraction field of a quotient domain.

| The residue field $\kappa(\mathfrak{p})$

Let B be a ring and $\mathfrak{p} \in \operatorname{Spec} B$. The **residue field** at $\mathfrak{p}$ is

$$\kappa(\mathfrak{p}) := \operatorname{Frac}(B/\mathfrak{p}).$$

It arises from a natural two-step factorization of $B \rightarrow \kappa(\mathfrak{p})$:

$$B \xrightarrow{\text{quotient}} B/\mathfrak{p} \xrightarrow{\text{localize}} \operatorname{Frac}(B/\mathfrak{p}) = \kappa(\mathfrak{p}).$$

Examples for $B = \mathbb{Z}$.

- $\mathfrak{p} = (0)$: $\mathbb{Z}/(0) = \mathbb{Z}$, $\kappa((0)) = \operatorname{Frac}(\mathbb{Z}) = \mathbb{Q}$.
- $\mathfrak{p} = (p)$: $\mathbb{Z}/(p) = \mathbb{F}_p$ is already a field, $\kappa((p)) = \mathbb{F}_p$.

These are exactly the fields whose spectra gave us the two fibers of $\pi : \operatorname{Spec} \mathbb{Z}[x] \rightarrow \operatorname{Spec} \mathbb{Z}$.

| $\mathrm{Spec} \kappa(y)$ as the scheme-theoretic point

The ring map $B \rightarrow \kappa(\mathfrak{p})$ induces a morphism of schemes

$$i_y : \mathrm{Spec} \kappa(y) \longrightarrow Y = \mathrm{Spec} B,$$

whose image is the single point $\{y\}$.

Intuition. $\mathrm{Spec} \kappa(y)$ is the *scheme-theoretic incarnation* of the point y . To locate y inside Y , the two steps do:

- *Quotient* $B \rightarrow B/\mathfrak{p}$: cuts out the closed subscheme $\overline{\{y\}}$, the closure of y .
- *Localize* $B/\mathfrak{p} \rightarrow \kappa(\mathfrak{p})$: pinpoints y as the generic point of $\overline{\{y\}}$.

In other words: a point $y \in Y$ is the same data as a morphism $\mathrm{Spec} \kappa(y) \rightarrow Y$. This is the correct way to *map a point into a scheme*.

| The tensor product and the affine fiber product

Let $\varphi : B \rightarrow A$ and $\psi : B \rightarrow C$ be ring homomorphisms. The **tensor product** $A \otimes_B C$ is generated by symbols $a \otimes c$ subject to B -bilinearity:

$$\varphi(b) a \otimes c = a \otimes \psi(b) c \quad \text{for all } b \in B.$$

Key fact. The fiber product of affine schemes over $\operatorname{Spec} B$ is:

$$\operatorname{Spec} A \times_{\operatorname{Spec} B} \operatorname{Spec} C = \operatorname{Spec}(A \otimes_B C).$$

Two examples we already know.

- $\mathbb{Z}[x] \otimes_{\mathbb{Z}} \mathbb{Q} \cong \mathbb{Q}[x].$
- $\mathbb{Z}[x] \otimes_{\mathbb{Z}} \mathbb{F}_p \cong \mathbb{F}_p[x].$

These are the fibers of π we computed at the start — now understood as tensor products.

The scheme-theoretic fiber

Let $f: X \rightarrow Y$ be a morphism of schemes, $y \in Y$ a point. In the affine case $X = \operatorname{Spec} A$, $Y = \operatorname{Spec} B$, f corresponds to $\varphi: B \rightarrow A$.

Definition. The **scheme-theoretic fiber** of f over y is the fiber product

$$X_y := X \times_Y \operatorname{Spec} \kappa(y) = \operatorname{Spec}(A \otimes_B \kappa(\mathfrak{p})).$$

What we still need to check. The set of prime ideals of $A \otimes_B \kappa(\mathfrak{p})$ should be exactly the set-theoretic fiber:

$$\{ \mathfrak{q} \in \operatorname{Spec} A \mid \varphi^{-1}(\mathfrak{q}) = \mathfrak{p} \}.$$

This is what we verify on the next slides using the two-step factorization $B \rightarrow B/\mathfrak{p} \rightarrow \kappa(\mathfrak{p})$.

Step 1 of 2: the quotient

The factorization $B \rightarrow B/\mathfrak{p} \rightarrow \kappa(\mathfrak{p})$ splits the computation of $A \otimes_B \kappa(\mathfrak{p})$ into two steps.

First step — tensor with $B/\mathfrak{p}$:

$$A \otimes_B B/\mathfrak{p} \cong A/\mathfrak{p}^e,$$

where $\mathfrak{p}^e := \varphi(\mathfrak{p}) \cdot A$ is the *extended ideal* of $\mathfrak{p}$ in A . (This follows from right-exactness of $\otimes$ applied to $\mathfrak{p} \rightarrow B \rightarrow B/\mathfrak{p} \rightarrow 0$.)

Geometric meaning. $\mathrm{Spec}(A/\mathfrak{p}^e) = f^{-1}(\overline{\{y\}})$, the scheme-theoretic preimage of the *closure* of y .

Which primes survive? The primes of $A/\mathfrak{p}^e$ are those $\mathfrak{q} \in \mathrm{Spec} A$ satisfying

$$\varphi^{-1}(\mathfrak{q}) \supseteq \mathfrak{p}.$$

This is too broad — it includes primes over proper specializations of $\mathfrak{p}$. The localization step will cut those out.

Step 2 of 2: the localization

Second step — localize $A/\mathfrak{p}^e$ at $S = \varphi(B \setminus \mathfrak{p})$:

$$S^{-1}(A/\mathfrak{p}^e) \cong (A/\mathfrak{p}^e) \otimes_{B/\mathfrak{p}} \kappa(\mathfrak{p}) \cong A \otimes_B \kappa(\mathfrak{p}).$$

Effect. Localization at S kills every prime $\mathfrak{q} \subset A/\mathfrak{p}^e$ that meets S , i.e. every $\mathfrak{q}$ for which $\mathfrak{q} \cap \varphi(B \setminus \mathfrak{p}) \neq \emptyset$, equivalently

$$\varphi^{-1}(\mathfrak{q}) \not\subseteq \mathfrak{p}.$$

Only the primes with $\varphi^{-1}(\mathfrak{q}) \subseteq \mathfrak{p}$ survive.

Geometric meaning. We are shrinking from the preimage of $\overline{\{y\}}$ down to the preimage of the generic point of $\overline{\{y\}}$ — which is y itself.

The two conditions together

A prime $\mathfrak{q} \in \operatorname{Spec} A$ survives both operations iff it satisfies both constraints simultaneously:

$$\varphi^{-1}(\mathfrak{q}) \supseteq \mathfrak{p} \quad \text{and} \quad \varphi^{-1}(\mathfrak{q}) \subseteq \mathfrak{p} \quad \implies \quad \varphi^{-1}(\mathfrak{q}) = \mathfrak{p}.$$

Operation	Algebra	Primes kept
Quotient	$A \rightarrow A/\mathfrak{p}^e$	$\varphi^{-1}(\mathfrak{q}) \supseteq \mathfrak{p}$
Localize	$A/\mathfrak{p}^e \rightarrow S^{-1}(A/\mathfrak{p}^e)$	$\varphi^{-1}(\mathfrak{q}) \subseteq \mathfrak{p}$
Combined	$A \rightarrow A \otimes_B \kappa(\mathfrak{p})$	$\varphi^{-1}(\mathfrak{q}) = \mathfrak{p}$

The underlying set of $\operatorname{Spec}(A \otimes_B \kappa(\mathfrak{p}))$ is the set-theoretic fiber $f^{-1}(y)$, now equipped with a natural scheme structure.

Back to $\operatorname{Spec} \mathbb{Z}[x]$: computing the fibers

Apply the machinery to $\varphi : \mathbb{Z} \hookrightarrow \mathbb{Z}[x]$.

Generic fiber over $(0) \in \operatorname{Spec} \mathbb{Z}$:

$$\kappa((0)) = \mathbb{Q}, \quad \mathbb{Z}[x] \otimes_{\mathbb{Z}} \mathbb{Q} \cong \mathbb{Q}[x].$$

The fiber is $\operatorname{Spec} \mathbb{Q}[x]$, the affine line over $\mathbb{Q}$.

Special fiber over $(p) \in \operatorname{Spec} \mathbb{Z}$:

$$\kappa((p)) = \mathbb{F}_p, \quad \mathbb{Z}[x] \otimes_{\mathbb{Z}} \mathbb{F}_p \cong \mathbb{F}_p[x].$$

The fiber is $\operatorname{Spec} \mathbb{F}_p[x]$, the affine line over $\mathbb{F}_p$.

The case split from Slide 3 was always this: each case corresponds to a fiber, and the *passage to $\mathbb{Q}[x]$ or $\mathbb{F}_p[x]$* is exactly the base change $\mathbb{Z}[x] \otimes_{\mathbb{Z}} \kappa(\mathfrak{p})$.

| The four types, revisited

Each prime of $\operatorname{Spec} \mathbb{Z}[x]$ lives in exactly one fiber, and plays a definite role within it.

Type	Prime	Fiber $\mathbb{Z}[x] \otimes_{\mathbb{Z}} \kappa(\cdot)$	Role in fiber
1	(0)	$\operatorname{Spec} \mathbb{Q}[x]$	Generic point
2	$(q(x)), q$ irred. prim.	$\operatorname{Spec} \mathbb{Q}[x]$	Closed point (q)
3	$(p), p$ prime	$\operatorname{Spec} \mathbb{F}_p[x]$	Generic point
4	$(p, q(x)), q$ irred. mod p	$\operatorname{Spec} \mathbb{F}_p[x]$	Closed point $(\bar{q})$

Types 1 and 2 belong to the *generic fiber*; Types 3 and 4 to a *special fiber over p* . The case split from the beginning was always a fiber-by-fiber analysis.

| Specialization: how fibers meet

A prime q' is a **specialization** of q if $q \subsetneq q'$, i.e. q' lies in the closure of q in the Zariski topology.

In $\text{Spec } \mathbb{Z}[x]$. A Type 2 prime $(q(x))$ specializes to a Type 4 prime $(p, \bar{q}(x))$ precisely when $\bar{q}(x) = q(x) \bmod p$ is still irreducible over $\mathbb{F}_p$.

- If $q(x)$ factors mod p , its specialization splits across *several* Type 4 primes — this is the geometric shadow of prime factorization in $\mathbb{F}_p[x]$.
- If $q(x)$ has a repeated factor mod p , the fiber over p is *ramified* — a phenomenon central to algebraic number theory.

The fiber product computes *which* primes land in each fiber;
specialization tells us how fibers relate to one another across $\text{Spec } \mathbb{Z}$.

Base change: reduction mod p

The fiber product $X \times_Y \operatorname{Spec} \kappa(y)$ is also called the **base change** of X along i_y . It turns a scheme over Y into a scheme over the single point $\kappa(y)$.

For $\operatorname{Spec} \mathbb{Z}[x]$. Given a polynomial $f \in \mathbb{Z}[x]$, the zero locus $V(f) \subset \operatorname{Spec} \mathbb{Z}[x]$ is a scheme over $\operatorname{Spec} \mathbb{Z}$. Its fibers are:

- $V(f) \times_{\operatorname{Spec} \mathbb{Z}} \operatorname{Spec} \mathbb{Q} = \operatorname{Spec} \mathbb{Q}[x]/(f)$ — roots of f over $\mathbb{Q}$.
- $V(f) \times_{\operatorname{Spec} \mathbb{Z}} \operatorname{Spec} \mathbb{F}_p = \operatorname{Spec} \mathbb{F}_p[x]/(\bar{f})$ — roots of $f \bmod p$.

Slogan. *Reduction mod p is base change to the special fiber.* The fiber product is the precise algebraic meaning of this operation.

| Two philosophies: charts versus algebra

Complex geometry. A complex manifold is built chart by chart.

- Cover X by open sets $U_\alpha \cong \mathbb{C}^n$.
- Write down transition functions $\varphi_{\alpha\beta}$ on overlaps.
- Define a bundle by local trivializations satisfying the cocycle condition on triple overlaps.
- Define a section by compatible local data on each chart.

The local picture is primary. The global picture is assembled from it.

Algebraic geometry. Start with a single global algebraic object — a ring, a graded ring, a sheaf.

- The local picture is *derived*: apply localization at a prime to zoom in near a point.
- No coordinates to choose, no transition functions to verify.

The global picture is primary. Localization hands you the local data for free.

| The structure sheaf: sections without coordinates

On a complex manifold X , the **structure sheaf** $\mathcal{O}_X$ assigns to each open $U \subset X$ the ring of holomorphic functions $\mathcal{O}_X(U)$, with restriction maps for $V \subset U$.

The same definition works on any scheme. On $\text{Spec } A$:

$$\mathcal{O}_X(D(f)) = A_f \quad (\text{localization of } A \text{ at } f).$$

No coordinates — the ring A is the global data; the local rings A_f are derived from it by localization.

A **sheaf** $\mathcal{F}$ on X is a rule assigning a module $\mathcal{F}(U)$ to each open U , compatible with restrictions, satisfying a gluing condition. It is the algebraic replacement for a holomorphic vector bundle — but defined globally, not chart by chart.

The key shift. In complex geometry a bundle is a collection of local pieces that agree. In algebraic geometry a sheaf is a single global object that localizes.

| Graded rings: algebra remembers scaling

A **graded ring** is a ring with a decomposition

$$S = S_0 \oplus S_1 \oplus S_2 \oplus \cdots, \quad S_i \cdot S_j \subset S_{i+j}.$$

An element of S_d is said to have *degree* d .

For a complex geometer. The grading encodes a $\mathbb{C}^*$ -action: an element $f \in S_d$ transforms as $f \mapsto \lambda^d f$ under $\lambda \in \mathbb{C}^*$. This is exactly the action of $\mathbb{C}^*$ on $\mathbb{C}^{n+1} \setminus \{0\}$ by scalar multiplication — the action whose quotient is $\mathbb{P}^n$.

Prototype. $S = k[x_0, \dots, x_n]$ with $\deg x_i = 1$. The degree- d piece S_d is spanned by monomials of total degree d . The ring encodes how homogeneous polynomials transform under scaling.

The point. Wherever a $\mathbb{C}^*$ -action appears geometrically, a graded ring appears algebraically. $\text{Proj}(S)$ is the algebraic way to take the quotient by that action.

Two graded algebras from a closed embedding

Let $Y = \operatorname{Spec} A$ and $X = V(I) \subset Y$ a closed subscheme. Two graded algebras arise naturally:

Associated graded $\operatorname{gr}_I(A) = \bigoplus_{n \geq 0} I^n / I^{n+1}$.

Elements are *leading terms* along X : a function $a \in I^n \setminus I^{n+1}$ vanishes to exact order n , and its image in I^n / I^{n+1} is its tangential part.

Geometrically: $\operatorname{Spec}(\operatorname{gr}_I(A))$ is the **normal cone** $C_X Y$.

Rees algebra $\operatorname{Rees}(I) = \bigoplus_{n \geq 0} I^n t^n \subset A[t]$.

Encodes the full filtration, not just the leading terms. Geometrically: $\operatorname{Proj}(\operatorname{Rees}(I))$ is the **blow-up** $\operatorname{Bl}_X Y$ — the scheme that separates directions of approach to X .

Punchline. A single closed embedding $X \hookrightarrow Y$ produces two canonical graded algebras. Applying Proj turns them into geometric objects — no coordinates chosen at any step.

Proj(S): projective space without coordinates

Let $S = \bigoplus_{n \geq 0} S_n$ be a graded ring. Define

$$\text{Proj}(S) := \{ \text{homogeneous prime ideals } \mathfrak{p} \subset S \mid \mathfrak{p} \not\supset S_+ \},$$

where $S_+ = \bigoplus_{n \geq 1} S_n$ is the *irrelevant ideal*.

The affine cover. For any homogeneous $f \in S_+$, the locus $D_+(f) = \{ \mathfrak{p} : f \notin \mathfrak{p} \}$ is an affine open:

$$D_+(f) \cong \text{Spec}(S_{(f)}), \quad S_{(f)} = \left\{ \frac{g}{f^n} : g \in S_n \right\}.$$

Example: $\mathbb{P}_k^1 = \text{Proj}(k[s, t])$.

- $D_+(s) \cong \text{Spec}(k[t/s]) = \mathbb{A}_k^1$.
- $D_+(t) \cong \text{Spec}(k[s/t]) = \mathbb{A}_k^1$.
- **Overlap:** $D_+(st) \cong \text{Spec}(k[t/s, (t/s)^{-1}]) = \mathbb{A}^1 \setminus \{0\}$.

The transition function $t/s \mapsto s/t$ is automatic from the algebra. *No coordinate patch was chosen.*

Why $\mathbb{P}^1$ is not affine — and why that is fine

Suppose $\mathbb{P}_k^1 = \operatorname{Spec} A$ for some ring A . Then $A = \mathcal{O}(\mathbb{P}^1) = H^0(\mathbb{P}^1, \mathcal{O})$.

But a global regular function on $\mathbb{P}^1$ is a rational function with no poles anywhere — hence a constant. So $A = k$, and $\operatorname{Spec} k$ is a single point. Contradiction.

What Proj does instead. The graded ring $k[s, t]$ is global data for $\mathbb{P}^1$, but it is *not* the ring of global functions — it is the ring of sections of all line bundles $\mathcal{O}(d)$ at once:

$$k[s, t]_d = H^0(\mathbb{P}^1, \mathcal{O}(d)).$$

The non-affineness of $\mathbb{P}^1$ reflects its *compactness* (no non-constant bounded functions). Proj is the correct algebraic notion of compactness — the analog of a compact complex manifold.

The direct image sheaf: one definition, no coordinates

Let $f: X \rightarrow Y$ be a morphism of schemes and $\mathcal{F}$ a sheaf on X . The **direct image** $f_*\mathcal{F}$ is the sheaf on Y defined by

$$(f_*\mathcal{F})(U) := \mathcal{F}(f^{-1}(U)) \quad \text{for every open } U \subset Y.$$

Restriction maps: $f^{-1}(V) \subset f^{-1}(U)$ for $V \subset U$, so restriction of $\mathcal{F}$ gives the required map. The sheaf axioms for $f_*\mathcal{F}$ follow immediately from those of $\mathcal{F}$ — since f^{-1} commutes with unions and intersections.

Compare with complex geometry. To define the "bundle of fiber cohomologies" $R^0 f_*\mathcal{L}$ on a holomorphic family, one needs harmonic theory, L^2 metrics, and Hodge theory to identify global sections fiber by fiber. Here: one line, no analysis, no metrics. The algebra does the same job globally.

| The affine case: restriction of scalars

Let $f: \operatorname{Spec} A \rightarrow \operatorname{Spec} B$ correspond to $\varphi: B \rightarrow A$, and let $\mathcal{F} = \widetilde{M}$ for an A -module M .

Claim. $f_*\widetilde{M} \cong \widetilde{M}_B$, where M_B is M viewed as a B -module via φ . *The pushforward is restriction of scalars — forget the A -structure, keep only the B -structure.*

The fiber, revisited. The fiber of $f_*\widetilde{M}$ at $\mathfrak{p} \in \operatorname{Spec} B$ is

$$(f_*\widetilde{M})|_{\mathfrak{p}} = M_B \otimes_B \kappa(\mathfrak{p}) = M \otimes_A \mathcal{O}_{X_{\mathfrak{p}}},$$

where $X_{\mathfrak{p}} = \operatorname{Spec}(A \otimes_B \kappa(\mathfrak{p}))$ is the scheme-theoretic fiber we constructed earlier.

The direct image closes the loop: the fiber of $f_*\mathcal{F}$ over a point y is precisely $\mathcal{F}$ restricted to the fiber product $X \times_Y \operatorname{Spec} \kappa(y)$.

Example: $\mathbb{P}^1$ -bundle over $\mathbb{A}^1$

Let $B = k[t]$, $X = \mathbb{P}_B^1 = \text{Proj}(B[s_0, s_1])$, and $f: X \rightarrow \text{Spec } B = \mathbb{A}^1$. Every fiber $f^{-1}(t_0) = \mathbb{P}_k^1$.

The direct image of $\mathcal{O}(d)$. The degree- d piece of the graded ring is

$$H^0(\mathbb{P}_B^1, \mathcal{O}(d)) = \bigoplus_{j=0}^d B \cdot s_0^j s_1^{d-j} \cong B^{d+1}.$$

Therefore $f_*\mathcal{O}(d) \cong \mathcal{O}_{\mathbb{A}^1}^{d+1}$: a free rank- $(d+1)$ sheaf on the base.

Checking the fiber. At any point $t_0 \in \mathbb{A}^1$:

$$(f_*\mathcal{O}(d))|_{t_0} = B^{d+1} \otimes_{k[t]} k = k^{d+1} = H^0(\mathbb{P}_k^1, \mathcal{O}(d)).$$

The direct image packages the cohomology of *every* fiber into a single algebraic object on the base. Tensoring with $\kappa(t_0)$ recovers any individual fiber — no integration, no L^2 theory required.

| The tautological line bundle $\mathcal{O}(1)$ on $\text{Proj}(S)$

On $\mathbb{P}^n = \text{Proj}(k[x_0, \dots, x_n])$, the sheaf $\mathcal{O}(d)$ is defined by its sections on basic opens:

$$\mathcal{O}(d)(D_+(f)) := \left\{ \frac{g}{f^m} : g \in S_{d+m \deg f} \right\}.$$

Geometrically: $\mathcal{O}(1)$ is the *hyperplane line bundle* — its global sections are exactly the linear forms $\sum a_i x_i$.

Key properties.

- $H^0(\mathbb{P}^n, \mathcal{O}(d)) = S_d$ for $d \geq 0$: global sections are degree- d homogeneous polynomials.
- $\mathcal{O}(1)$ is *ample*: its powers $\mathcal{O}(d)$ embed $\mathbb{P}^n$ into projective space.
- For any $\text{Proj}(S)$, the same construction gives a tautological $\mathcal{O}(1)$ encoding the grading.

The same idea applies to $\text{Proj}(\text{Rees}(I))$ and $\text{Proj}(\text{gr}_I(A))$ — the blow-up and the normal cone both carry natural line bundles inherited from the grading.

The extended Rees algebra

The standard Rees algebra $\bigoplus_{n \geq 0} I^n t^n$ lives over $\mathbb{A}_t^1$, but has no well-behaved fiber at $t = \infty$. The fix: extend the grading to all integers.

Definition. Set $u = t^{-1}$ (the deformation parameter). The **extended Rees algebra** is

$$R'(I) = \bigoplus_{n \in \mathbb{Z}} I^n u^{-n} \subset A[u, u^{-1}], \quad I^n := A \text{ for } n \leq 0.$$

Two structural facts.

- **At $u = 0$:** $R'(I)/(u) \cong \operatorname{gr}_I(A) = \bigoplus_{n \geq 0} I^n / I^{n+1}$.
- **Away from $u = 0$:** $R'(I)[u^{-1}] = A[u, u^{-1}]$ — the filtration by I vanishes once u is invertible.

Note. Since $I \subsetneq A$ is proper, $u^{-1} = t \notin R'(I)$: the degree-1 piece is $I \cdot u^{-1}$, not $A \cdot u^{-1}$. So $R'(I)$ is an $A[u]$ -algebra — the family lives over $\mathbb{A}_u^1$, not $\mathbb{G}_m$.

| The flat family over $\mathbb{A}^1$

The inclusion $A[u] \hookrightarrow R'(I)$ makes

$$\mathcal{M} = \operatorname{Spec}(R'(I)) \longrightarrow Y \times \mathbb{A}_u^1 = \operatorname{Spec}(A[u])$$

a **flat** morphism — the deformation of Y to the normal cone of X .

Locus	Fiber	Ring
$u \neq 0$	$Y = \operatorname{Spec}(A)$ (constant)	$R'(I)[u^{-1}] = A[u, u^{-1}]$
$u = 0$	$C_X Y$ (normal cone)	$R'(I)/(u) \cong \operatorname{gr}_I(A)$

Why flatness matters. A flat family preserves the cycle class of its fibers. Consequently $[Y]$ and $[C_X Y]$ are *rationaly equivalent* cycles — they are fibers of a flat family over $\mathbb{P}^1$, hence differ by a boundary in the Chow group. This is precisely what makes the specialization map $\sigma : A_k(Y) \rightarrow A_k(C_X Y)$ well-defined.

| The special fiber sees only leading terms

Let $a \in I^n \setminus I^{n+1}$: we say a has **order n along X** .

- **In $R'(I)$:** a appears at u -degree $-k$ for $k = 0, 1, \dots, n$ (as $a \cdot u^{-k} \in I^k \cdot u^{-k}$), but *not* at degree $-(n+1)$, since $a \notin I^{n+1}$. So $-n$ is the most negative degree at which a appears.
- **Setting $u = 0$:** the degree- $(-k)$ piece of $R'(I)/(u)$ is I^k/I^{k+1} . For $k < n$: $a \in I^k$ but the quotient kills it (it is not in $I^k \setminus I^{k+1}$). At $k = n$: a maps to $[a] \in I^n/I^{n+1}$ — its *leading term* — which is nonzero.

$C_X Y = \operatorname{Spec}(\operatorname{gr}_I(A))$ is the **scheme of leading terms**: for each $a \in A$, only the lowest-order part of a along I survives in the special fiber. It is the algebraic tangent cone to Y along X .

| Normal cone, blow-up, and the deformation picture

Two canonical schemes arise from the closed embedding $X \hookrightarrow Y$:

- **Normal cone:** $C_X Y = \operatorname{Spec}(\operatorname{gr}_I(A))$. The grading gives a $\mathbb{G}_m$ -action on $C_X Y$ over X — this is what makes it a *cone*.
- **Blow-up:** $\operatorname{Bl}_X Y = \operatorname{Proj}(\operatorname{Rees}(I))$, with exceptional divisor

$$E = \operatorname{Proj}(\operatorname{gr}_I(A)) = \mathbb{P}(C_X Y).$$

Outside E , the blow-up is isomorphic to $Y \setminus X$.

The interpolation. Let $M^\circ = \mathcal{M} \setminus \tilde{Y}$ (remove the blow-up component from the special fiber). Then $M^\circ \rightarrow \mathbb{P}^1$ is a flat family:

fiber over $u \neq 0$: $X \hookrightarrow Y$, fiber over $u = 0$: $X \hookrightarrow C_X Y$ (zero section).

The normal cone is the limit of Y as we zoom toward X — the first-order approximation to the embedding.

| The ideal sheaf and the conormal sheaf

Let $i: Y = V(I) \hookrightarrow X = \operatorname{Spec}(A)$ be a closed embedding.

The ideal $I \subset A$ defines a sheaf $\mathcal{I}$ on X . Its restriction to Y is the **conormal sheaf**:

$$\mathcal{I}/\mathcal{I}^2 = I/I^2 \quad (\text{an } A/I\text{-module, i.e. a sheaf on } Y).$$

Why I/I^2 ? Elements of I^2 vanish to second order along Y — they carry no first-order (linear) information. Modding out by I^2 keeps only the *linear part* of each defining equation. This is the algebraic analog of taking the derivative of a vanishing condition.

Example. $X = \mathbb{A}^n$, $Y = V(x_1, \dots, x_r)$ a coordinate subspace. Then $I = (x_1, \dots, x_r)$ and

$$I/I^2 \cong (A/I)^{\oplus r}, \quad \text{with basis } \bar{x}_1, \dots, \bar{x}_r.$$

These are the r normal directions to Y in X , linearized.

The conormal exact sequence and the normal bundle

Theorem. For $i: Y \hookrightarrow X$ a closed embedding, there is an exact sequence on Y :

$$0 \longrightarrow \mathcal{I}/\mathcal{I}^2 \longrightarrow \Omega_X^1|_Y \longrightarrow \Omega_Y^1 \longrightarrow 0.$$

Proof sketch. The Leibniz rule gives $d(fg) = f dg + g df$, so $d(I^2) \subset I \cdot \Omega_A^1$. Hence $d: I \rightarrow \Omega_A^1$ descends to a well-defined map $I/I^2 \rightarrow \Omega_X^1|_Y$. The cokernel is Ω_Y^1 by definition of the relative Kähler differentials.

The normal bundle. Define $N_{Y/X} = (I/I^2)^\vee = \mathcal{H}om_{A/I}(I/I^2, A/I)$. When I/I^2 is locally free — i.e. the embedding is *regular* — we have $\mathrm{gr}_I(A) \cong \mathrm{Sym}(I/I^2)$, so:

$$C_X Y = \mathrm{Tot}(N_{Y/X}).$$

The deformation family of the previous section is exactly *deforming* Y *into the total space of its own normal bundle*.

Examples of the conormal sequence

Smooth hypersurface. Let $Y = V(f) \subset X = \mathbb{A}^n$, f squarefree. Then $I = (f)$, $I/I^2 \cong \mathcal{O}_Y$ (rank 1, generated by $\bar{f}$), and the sequence is:

$$0 \longrightarrow \mathcal{O}_Y \xrightarrow{\cdot df|_Y} \Omega_{\mathbb{A}^n}^1|_Y \longrightarrow \Omega_Y^1 \longrightarrow 0.$$

The first map sends $1 \mapsto df|_Y$. The normal bundle $N_{Y/X} = \mathcal{O}_Y$ is trivial — one independent normal direction, generated by the gradient of f .

Complete intersection. Let $Y = V(f_1, \dots, f_r) \subset X$ with $(f_1, \dots, f_r)$ a regular sequence. Then I/I^2 is locally free of rank r with basis $\bar{f}_1, \dots, \bar{f}_r$:

$$0 \longrightarrow \mathcal{O}_Y^r \longrightarrow \Omega_X^1|_Y \longrightarrow \Omega_Y^1 \longrightarrow 0.$$

The normal bundle $N_{Y/X}$ has rank r — one normal direction per equation.

Taking determinants: the adjunction formula

Key algebraic fact. For a short exact sequence of locally free sheaves $0 \rightarrow \mathcal{A} \rightarrow \mathcal{B} \rightarrow \mathcal{C} \rightarrow 0$:

$$\det \mathcal{B} = \det \mathcal{A} \otimes \det \mathcal{C}, \quad \det \mathcal{E} := \bigwedge^{\text{rank } \mathcal{E}} \mathcal{E}.$$

Apply to the conormal sequence. Taking $\det$ of

$0 \rightarrow \mathcal{I}/\mathcal{I}^2 \rightarrow \Omega_X^1|_Y \rightarrow \Omega_Y^1 \rightarrow 0$ gives

$K_X|_Y = \det(\mathcal{I}/\mathcal{I}^2) \otimes K_Y = (\det N_{Y/X})^* \otimes K_Y$. Rearranging:

$$K_Y = i^* K_X \otimes \det N_{Y/X}.$$

What comes next. This derivation was purely formal. The complex-analytic proof (Poincaré residue) illuminates the geometry; the algebraic proof (Koszul complex) will be our gateway to Ext and Tor.

Adjunction: forms with poles

Setup. Let $Y \subset X$ be a smooth divisor in a complex manifold of dimension n , defined locally by $\{f = 0\}$. The line bundle $\mathcal{O}(Y)$ has Y as its zero locus, so $\mathcal{O}(Y)|_Y \cong N_{Y/X}$.

The adjunction formula (to be proved):

$$K_Y = (K_X \otimes \mathcal{O}(Y))|_Y = K_X|_Y \otimes N_{Y/X}.$$

The key observation.

- $K_X \otimes \mathcal{O}(Y)$ = holomorphic n -forms on X with at most a *simple pole* along Y .
- K_Y = holomorphic $(n - 1)$ -forms on Y .

Question. How does a simple pole along a codimension-1 submanifold produce a form of one lower degree on that submanifold?

The Poincaré residue map

Let $Y = \{f = 0\} \subset X$ and let ω be a holomorphic n -form on $X \setminus Y$ with at most a simple pole along Y . Locally, one can always write

$$\omega = \frac{df}{f} \wedge \alpha + \beta,$$

where α is a holomorphic $(n-1)$ -form and β is holomorphic.

Definition. The **Poincaré residue** of ω along Y is $\text{Res}_Y(\omega) := \alpha|_Y \in K_Y$. This is independent of the choice of local equation f and of the splitting: replacing f by gf shifts α by a form vanishing on Y .

Global statement. The residue assembles into a sheaf map $\text{Res} : K_X \otimes \mathcal{O}(Y) \rightarrow i_* K_Y$, defined globally, with no coordinate choices.

The residue exact sequence

The residue map fits into a short exact sequence on X :

$$0 \longrightarrow K_X \xrightarrow{\cdot f} K_X \otimes \mathcal{O}(Y) \xrightarrow{\text{Res}} i_* K_Y \longrightarrow 0.$$

- **Left map $\cdot f$:** multiplying a holomorphic n -form by $f \in \mathcal{O}(Y)$ cancels the simple pole. Its image is exactly the kernel of Res .
- **Surjectivity:** given any $(n-1)$ -form α on Y , extend it locally and form $\frac{df}{f} \wedge \alpha$, which maps to $\alpha|_Y$ under Res .

Adjunction follows. Restricting to Y , the residue gives an isomorphism $K_X(Y)|_Y \xrightarrow{\sim} K_Y$. Since $K_X(Y)|_Y = K_X|_Y \otimes \mathcal{O}(Y)|_Y = K_X|_Y \otimes N_{Y/X}$:

$$K_Y = K_X|_Y \otimes N_{Y/X}. \quad \square$$

| What the proof really used

The residue proof had two ingredients:

- **One equation** f , giving a length-1 free resolution of $\mathcal{O}_Y$:

$$0 \longrightarrow \mathcal{O}_X(-Y) \xrightarrow{f} \mathcal{O}_X \longrightarrow \mathcal{O}_Y \longrightarrow 0.$$

- **Applying** $\mathcal{H}om(-, K_X)$ and reading off the cokernel, which turned out to be $K_X|_Y \otimes N_{Y/X} = K_Y$.

For a complete intersection $Y = V(f_1, \dots, f_r)$: one equation is not enough. We need r equations and a longer free resolution — the **Koszul complex** — to carry out the same strategy.

The algebraic path. Apply $\mathcal{H}om(-, K_X)$ to the Koszul complex and read off $\mathcal{E}xt^r(\mathcal{O}_Y, K_X) = K_X|_Y \otimes \det N_{Y/X} = K_Y$. Carrying this out forces us to define Ext .

The Koszul complex

Hypersurface (length 1). For $f_1 \in A$ a non-zero-divisor, the resolution $0 \rightarrow A \xrightarrow{f_1} A \rightarrow A/(f_1) \rightarrow 0$ is free of length 1.

Regular sequence (length r). For a regular sequence $f_1, \dots, f_r$, the **Koszul complex** $K^\bullet(f_1, \dots, f_r)$ has terms $\bigwedge^k A^r$ in degree k , with differential

$$d(e_{i_1} \wedge \cdots \wedge e_{i_k}) = \sum_{j=1}^k (-1)^{j-1} f_{i_j} e_{i_1} \wedge \cdots \wedge \widehat{e_{i_j}} \wedge \cdots \wedge e_{i_k}.$$

Key theorem. $(f_1, \dots, f_r)$ regular $\iff K^\bullet$ is a free resolution of A/I :

$$0 \longrightarrow \bigwedge^r A^r \longrightarrow \cdots \longrightarrow A^r \longrightarrow A \longrightarrow A/I \longrightarrow 0.$$

The top term $\bigwedge^r A^r \cong A$ is free of rank 1 — this produces the single $\det N_{Y/X}$ twist in adjunction.

Applying $\mathcal{H}om(-, K_X)$: adjunction algebraically

Apply $\mathcal{H}om(-, K_X)$ to the hypersurface resolution $0 \rightarrow \mathcal{O}_X(-Y) \xrightarrow{f} \mathcal{O}_X \rightarrow 0$. Since $\mathcal{H}om(\mathcal{O}_X(-Y), K_X) = K_X(Y)$, the dual complex is:

$$0 \longrightarrow K_X \xrightarrow{f} K_X(Y) \longrightarrow 0.$$

- **Degree 0:** $\mathcal{E}xt^0(\mathcal{O}_Y, K_X) = \ker(\cdot f) = 0$ (K_X is torsion-free, f is a non-zero-divisor).
- **Degree 1:** $\mathcal{E}xt^1(\mathcal{O}_Y, K_X) = \operatorname{coker}(\cdot f) = K_X(Y)|_Y = K_X|_Y \otimes N_{Y/X}$.

The Koszul computation gives $\mathcal{E}xt^1(\mathcal{O}_Y, K_X) = K_X|_Y \otimes N_{Y/X} = K_Y$ — the adjunction formula, with no black box: just the cokernel of $\cdot f$.

| What is Ext?

The functor $\text{Hom}_A(M, -)$ is *left-exact*: given $0 \rightarrow N' \rightarrow N \rightarrow N'' \rightarrow 0$ exact, we get $0 \rightarrow \text{Hom}(M, N') \rightarrow \text{Hom}(M, N) \rightarrow \text{Hom}(M, N'')$ exact, but the last map need not be surjective.

Definition. Take a free resolution $\cdots \rightarrow P_1 \rightarrow P_0 \twoheadrightarrow M$ and apply $\text{Hom}(-, N)$:

$$0 \rightarrow \text{Hom}(P_0, N) \rightarrow \text{Hom}(P_1, N) \rightarrow \cdots$$

Then $\text{Ext}_A^i(M, N) = H^i$ of this complex. The result is independent of the resolution chosen.

Key properties.

- $\text{Ext}^0(M, N) = \text{Hom}(M, N)$.
- $\text{Ext}^i(A, N) = 0$ for $i > 0$ (free modules cause no failure).
- $\text{Ext}^i(M, N) = 0$ for all $i > \text{pd}(M)$.
- The Koszul complex provides an *explicit* resolution — making Ext directly computable.

What is Tor , and what is flatness?

The functor $M \otimes_A -$ is *right-exact*: given $0 \rightarrow N' \rightarrow N \rightarrow N'' \rightarrow 0$, we get $M \otimes N' \rightarrow M \otimes N \rightarrow M \otimes N'' \rightarrow 0$ exact, but the first map need not be injective.

Definition. Take a free resolution $\cdots \rightarrow P_1 \rightarrow P_0 \twoheadrightarrow M$ and apply $- \otimes_A N$:

$$\cdots \rightarrow P_1 \otimes N \rightarrow P_0 \otimes N \rightarrow 0.$$

Then $\text{Tor}_i^A(M, N) = H_i$ of this complex.

Flatness. M is **flat** over $A \iff \text{Tor}_i^A(M, N) = 0$ for all $i > 0$ and all N .

Connection to the deformation family. $R'(I)$ is flat over $A[u]$ because $\text{Tor}_i^{A[u]}(R'(I), N) = 0$ for $i > 0$: tensoring with $R'(I)$ stays exact. This is why the fiber over $u = 0$ is the honest normal cone, not a derived artifact — flatness guarantees no higher corrections.

| The dualizing sheaf $K_X = \Omega_X^n$

For a smooth variety X of dimension n , the **dualizing sheaf** is

$$K_X = \det \Omega_X^1 = \Omega_X^n = \bigwedge^n \Omega_X^1,$$

the algebraic canonical bundle — the same Ω_X^1 from the conormal sequence, now taken to the top exterior power.

Example: $K_{\mathbb{P}^n} = \mathcal{O}(-n-1)$. The **Euler exact sequence** on $\mathbb{P}^n$ is

$$0 \longrightarrow \Omega_{\mathbb{P}^n}^1 \longrightarrow \mathcal{O}(-1)^{\oplus(n+1)} \longrightarrow \mathcal{O} \longrightarrow 0.$$

Taking $\det$ (using multiplicativity from slide 37):

$$K_{\mathbb{P}^n} = \det(\mathcal{O}(-1)^{n+1}) = \mathcal{O}(-(n+1)).$$

For $n = 1$: $K_{\mathbb{P}^1} = \mathcal{O}(-2)$ — the exponent in the symmetry $H^1(\mathbb{P}^1, \mathcal{O}(d)) \cong H^0(\mathbb{P}^1, \mathcal{O}(-2-d))^\vee$ from slide 48. The dualizing sheaf was present all along.

The trace map: $H^n(X, K_X) \cong k$

For a smooth projective variety X of dimension n over k , there is a canonical isomorphism

$$\mathrm{tr} : H^n(X, K_X) \xrightarrow{\sim} k.$$

This is the **fundamental class** of X — the algebraic analog of integrating an n -form over a compact oriented manifold.

Computation on $\mathbb{P}^1$. With the standard cover $\{U_s, U_t\}$, the Čech complex for $K_{\mathbb{P}^1} = \mathcal{O}(-2)$ has $H^0 = 0$ (no global sections of a negative bundle) and

$$H^1(\mathbb{P}^1, \mathcal{O}(-2)) \cong k,$$

generated by the Čech 1-cocycle $\frac{1}{st}$ on $U_s \cap U_t$.

Intuition. The cocycle $\frac{1}{st}$ has simple poles at $s = 0$ and $t = 0$ — the algebraic shadow of $\frac{ds}{s} = -\frac{dt}{t}$. It cannot be split as a difference of sections on U_s and U_t , so it represents a nontrivial class: the fundamental class of $\mathbb{P}^1$.

The Serre duality pairing

Let X be smooth projective of dimension n , $\mathcal{F}$ locally free. The **cup product** gives a bilinear pairing:

$$H^i(X, \mathcal{F}) \times H^{n-i}(X, K_X \otimes \mathcal{F}^\vee) \longrightarrow H^n(X, K_X) \xrightarrow{\text{tr}} k.$$

Via Čech cocycles. A p -cocycle α for $\mathcal{F}$ and a $(n-p)$ -cocycle β for $K_X \otimes \mathcal{F}^\vee$ pair on (n) -fold overlaps via the contraction $\mathcal{F} \otimes (K_X \otimes \mathcal{F}^\vee) \rightarrow K_X$. The result is an n -cocycle for K_X , then sent to k by tr .

Properties.

- Natural in $\mathcal{F}$ — no extra choices beyond the trace map.
- For $i = 0$: pairs global sections of $\mathcal{F}$ against $H^n(K_X \otimes \mathcal{F}^\vee)$.
- For $i = n$: pairs $H^n(\mathcal{F})$ against global sections of $K_X \otimes \mathcal{F}^\vee$.

Serre duality

Theorem (Serre). Let X be smooth projective of dimension n over k , $\mathcal{F}$ locally free. The pairing of slide 52 is **perfect**:

$$H^i(X, \mathcal{F}) \cong H^{n-i}(X, K_X \otimes \mathcal{F}^\vee)^\vee.$$

Consequences.

- $H^i(X, \mathcal{F}) = 0$ for $i > n$ — cohomology vanishes above dimension.
- $\chi(X, \mathcal{F}) = (-1)^n \chi(X, K_X \otimes \mathcal{F}^\vee)$ — symmetry of the Euler characteristic.

Verify on $\mathbb{P}^1$. $K_{\mathbb{P}^1} \otimes \mathcal{O}(d)^\vee = \mathcal{O}(-2-d)$, so duality predicts

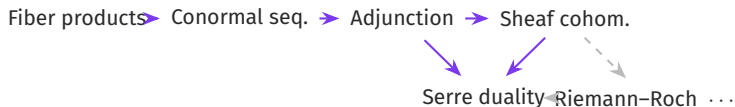
$$H^1(\mathbb{P}^1, \mathcal{O}(d)) \cong H^0(\mathbb{P}^1, \mathcal{O}(-2-d))^\vee.$$

This matches slide 48 exactly. The symmetry we computed there was Serre duality.

| The full arc: local to global

Adjunction is a *local* statement: the conormal sequence gives $K_Y = K_X|_Y \otimes \det N_{Y/X}$ — a pointwise identity of line bundles on Y .

Serre duality is the *global* counterpart: it pairs cohomology groups across the whole variety, with K_X as the dualizing object in both.



The residue exact sequence + long exact cohomology sequence + Serre duality on X and Y together imply **Riemann–Roch for divisors** — the natural next destination.